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Guidelines for use of the random positioning machine as a reduced-gravity analog

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With expanding spaceflight mission plans to the Moon, Mars, and beyond, investigating biological phenomena in reduced-gravity environments is a critical need. Ground-based spaceflight-analog platforms, such as the two-frame Random Positioning Machine (RPM), represent an alternative to operationally challenging spaceflight-based experiments. We developed a kinematic model to define the acceleration and gravitational level experienced by a sample at the center of an RPM. In comparing this to the RPM's performance with accelerometers and motion capture sensors, we validated the RPM and defined optimized performance parameters for time-averaged microgravity and reduced-gravity simulation (rotation rate: outer frame > 30 deg/s, inner frame > 20 deg/s slower; sample size: <10–15 cm from the center depending on rotation rate; experimental duration: >25 min). We also present the validated designs for an RPM capable of simulating microgravity, lunar gravity, Martian gravity, and hypergravity simultaneously. Finally, we identify "pole bias" as a critical limitation for two-frame RPMs: when the outer frame is aligned vertically, which occurs twice during each rotation, the gravity vector is aligned with a sample and generates two poles. At higher rotation rates, this significantly reduces the fidelity of an RPM's reduced-gravity simulation, and we thus introduce the design and validation for a novel three-frame RPM that can mitigate pole bias. We validated the RPM platform, optimized experimental design guidelines, and identified novel RPM designs for microgravity and reduced-gravity investigations.

Keywords Microgravity, Spaceflight analog, Validation, Random positioning machine (RPM), Guidelines, Partial gravity

With upcoming spaceflight missions poised to return to the Moon and enable interplanetary travel, characterizing the impacts of microgravity remains a critical challenge. For example, investigations to prevent the biological effects of reduced gravity from spaceflight travel range from the musculoskeletal, neurovestibular, immunological, cardiovascular systems, and more[1–4]. Material physical and chemical properties also differ significantly in the microgravity environment due to increased diffusion and surface tension but decreased sedimentation and buoyancy-driven convection[5,6]. This has major implications for pharmaceutical production, bioprinting, biotechnology-hardware design, and more in spaceflight[7,8]. Fully characterizing biological and physical science properties in reduced gravity therefore remains an active area of investigation for spaceflight research.

Yet, there are significant logistical limitations in conducting spaceflight experiments in low Earth Orbit (LEO) and beyond. Spaceflight studies are costly and operationally challenging, with restrictive mass, power, and volume limitations. Experiments in the partial gravity environments of the Moon or Mars are even more limited to execute. This has resulted in low experimental volumes, throughput, and sample sizes, all limiting the reproducibility of spaceflight-based experiments.

Hypogravity-analog platforms have therefore increased in popularity due to their greater accessibility and variations in microgravity-duration, allowing for customized experimental designs[9]. Short term (seconds to minutes) weightlessness can be generated terrestrially using drop towers, parabolic flights, and sounding rockets, which have been extensively reviewed elsewhere[10]. Longer-term ground-based reduced-gravity simulators

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include 1D-3D clinostats, the random positioning machine (RPM), and diamagnetic levitation systems which have also been discussed elsewhere[11,12].

Briefly, clinostats rotate around 1 or 2-axes with constant slow-rotation (1–2 rotations/min) or fast-rotation (50–100 rotations/min) modes[9,13]. The former establishes a central region where centrifugal forces are limited but is susceptible to mechanosensation in systems that can detect gravitational changes quickly, e.g., plant gravitropism[14,15]. Indeed, it has been shown that low-speed clinorotation (1–2 deg/sec) induces signs of gravitropism, as opposed to microgravity simulation, in plant root growth[16]. Conversely, fast rotation in a clinostat can achieve stable rotation state around a central sample, enabling studies of sedimentation in fluid or semisolid mediums. Higher rotation rates can also introduce adverse shear forces on the system[17]. Significant differences have been demonstrated between clinostat-based studies and true spaceflight experiments due to the clinostat's constant rotation rate and direction, limiting their use and applications[14].

Recent advancements in the control of 3D clinostats were introduced in Japan during the 1990's. Hoson and colleagues developed a platform where the rotational direction of the clinostat would randomly change every 30–60 s within a −2 to 2 rotations/min range, representing the first RPM and achieving improved fidelity for time-averaged microgravity simulation[14,18]. New control systems have also been developed, for example: generating a map of the inner- and outer-frame gravity vector orientations to develop a reference trajectory for the rotation[19], or "partial gravity" modes to simulate lunar or Martian gravity[20]. Incubator-compatible RPMs have further enabled cell-culture studies with the platform, allowing for expanded research on cellular mechanistic biology, tissue engineering, cancer pharmacology, and more[19,21,22]. The RPM (illustrated in Fig. 1A) has thus become an important tool for conducting experiments in hypogravity.

Despite this increased use of RPMs as a research platform, a full characterization of its performance and particularly its limitations is lacking[23,24]. For example, changing the rotation velocity impacts not only the time to reach hypogravity but also introduces additional forces. Centripetal, Coriolis, and Euler forces all act on a sample pulling it either towards the radial center, perpendicular to its relative velocity, or perpendicular to the angular acceleration and sample position vector, respectively. This also contributes to shear stresses acting parallel to a sample's surface, for example within the cell fluid medium, thus affecting cellular activation, survival, and signaling[25,26]. Each of these general factors, along with individual aspects of study design such as viscosity, cell-cell interactions, and cell-division rate, must be considered when designing a trial for an RPM. Theoretically, the equations governing simulated reduced gravity levels by averaging out the *G*-vector only applies in singular point, and thus, a principal limit in sample size can be estimated. Wuest and colleagues previously explored how sample distance from the center of rotation (COR) affects worst-case centrifugal and tangential acceleration, but a physical validation of this to outline sample size limits for different combinations of different rotation rates is yet unknown[23].

Here we characterize the RPM's kinematic performance by developing a defined mathematical model, compare and validate it experimentally using motion capture, and finally, define limitations relating to rotational velocity, the physical size of the sample, and experimental duration, with the aim of developing optimized experimental design guidelines for use by spaceflight scientists.

Methods

Defining the kinematics model

We first define the coordinate system for the kinematics model to describe the inner-frame and outer-frame RPM rotation, as well as sample points away from the COR. Three distinct coordinate systems are defined: (1) the global static frame ($_G$), (2) the outer frame of the RPM ($_O$) which rotates around the x_G-axis with angular velocity ω_O, and (3) the inner frame of the RPM ($_I$) which rotates around the y_O-axis with angular velocity ω_I (Fig. 1B). To switch between the 3 coordinate systems, rotational matrices are defined in Eqs. (1) and (2) for rotation about the x- and y-axis, respectively. The angle σ indicates the angle of rotation of the outer frame about x_O, and θ indicates the angle of the inner frame relative to the outer frame.

$$Rx = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\sigma & -\sin\sigma \\ 0 & \sin\sigma & \cos\sigma \end{bmatrix} \tag{1}$$

$$Ry = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \tag{2}$$

We then define point P as an arbitrary point on the platform at the inner frame. Its acceleration through the dynamical system described above is shown in Eq. (3), where $\alpha_{G,P}$ is point P's acceleration in terms of the global frame, α_G is acceleration of the global frame (in this case -9.81 m/s^2 due to Earth's gravity), $\alpha_{O,P}$ is relative acceleration between point P and the outer frame, $\alpha_{centripetal}$ is centripetal acceleration, $\alpha_{Coriolis}$ is Coriolis acceleration, and α_{Euler} is Euler acceleration. Additional definitions for the latter terms are shown in Supplementary Equations (S1) through (S3). Point P's relative acceleration to the outer frame $\alpha_{O,P}$ is shown in Eq. (4), where α_I is acceleration of the inner frame, and $\alpha_{O,I}$ is relative acceleration between point P and the inner frame. Combining Eqs. (3) and (4) defines the complete description of the acceleration at an arbitrary point P on the RPM based on the two frames' rotational rates and change in rotational rate, as shown in Eq. (5). Switching between coordinate systems can be done using the rotational matrices defined previously.

$$\alpha_{G,P} = \alpha_G + \alpha_{O,P} + \alpha_{centripetal} + \alpha_{Coriolis} + \alpha_{Euler} \tag{3}$$

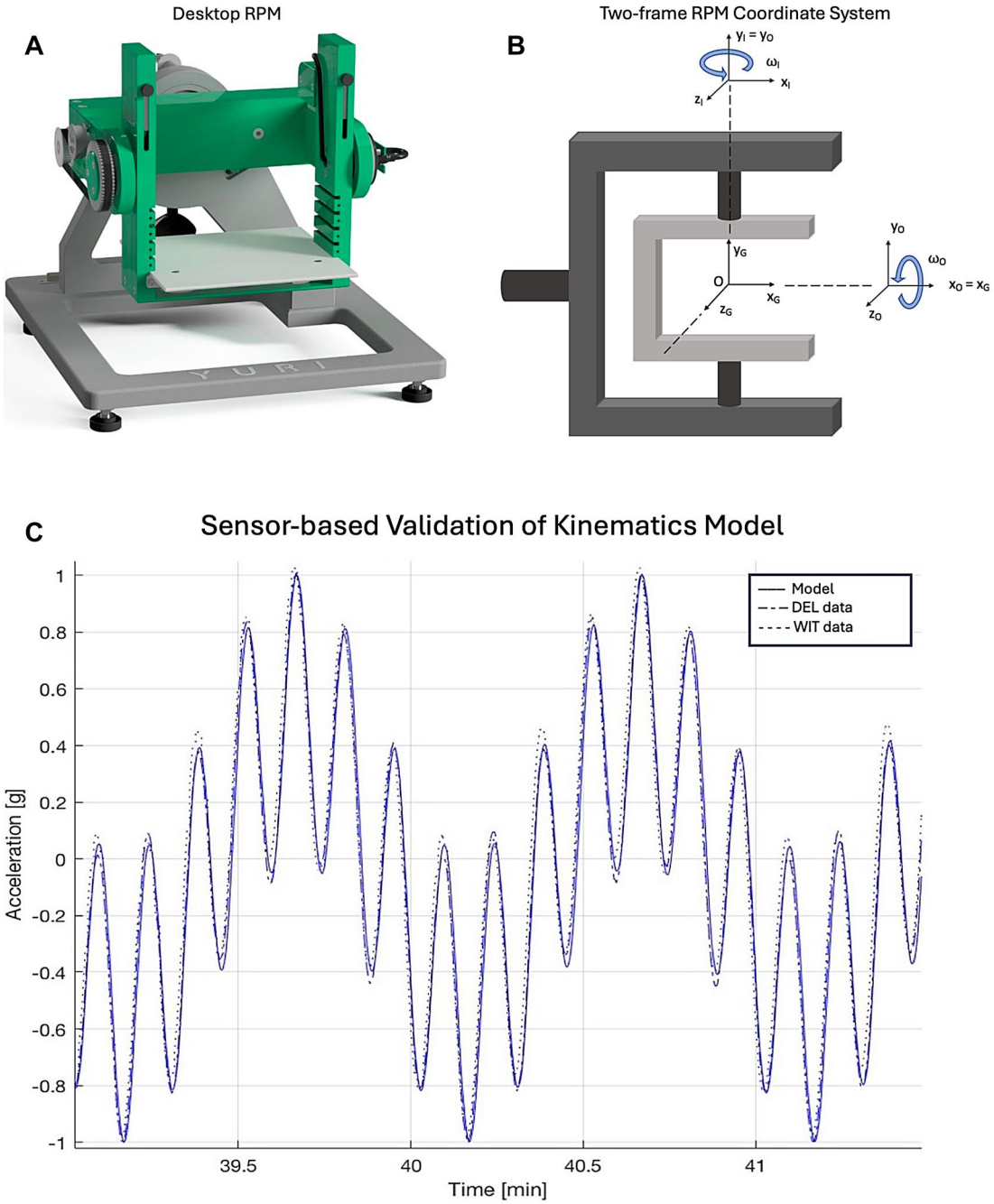


Fig. 1. Validation of the kinematics model for the random positioning machine (RPM). (**A**) Schematic of the RPM, Yuri Gravity (permission obtained). (**B**) Reference frame and adopted coordinates for the kinematics model: global static frame (G), outer frame coordinate system (o), and inner frame coordinator system (l). (**C**) Time-averaged acceleration (x-direction) of the corner point of an RPM (ω_O = 23.98 deg/s and ω_I = 17.966 deg/s). Comparing the kinematics model to Inertial Measurement Unit (IMU) Delsys sensor (DEL) data and WitMotion sensor (WIT) data.

$$\alpha_{O,P} = \alpha_I + \alpha_{O,I} + \alpha_{\text{centripetal}} + \alpha_{\text{Coriolis}} + \alpha_{\text{Euler}} \tag{4}$$

$$\begin{aligned}\alpha_{G,P} = \alpha_G + [\omega_{O,I} \times (\omega_{O,I} \times \overline{P}_{O,P})]_{G,O} + [\omega_{G,O} \times (\omega_{G,O} \times \overline{P}_{G,I})] \\ + 2\omega_{G,O}[\omega_{O,I} \times \overline{P}_{O,P}]_{G,O} + [\dot{\omega}_{O,I} \times \overline{P}_{O,P}]_{G,O} + [\dot{\omega}_{G,O} \times \overline{P}_{G,I}]\end{aligned} \tag{5}$$

To model the resulting *g*-level for the point P on the RPM, the time-averaged accelerations are calculated at each time step in each direction individually as shown in Eq. (6) and combined to a mean acceleration μ_a as shown in Eq. (7) for the *N*th time step. This method was used for both the kinematics model and the experimental data.

$$\alpha_{\text{time-average}}(\text{N}) = \begin{bmatrix} \alpha_{x,time-average} \\ \alpha_{y,time-average} \\ \alpha_{z,time-average} \end{bmatrix} = \frac{1}{N}\sum_{i=1}^{N} a_i \tag{6}$$

$$\mu_a = \sqrt{\alpha_{x,time-average}{}^2 + \alpha_{y,time-average}{}^2 + \alpha_{z,time-average}{}^2} \tag{7}$$

When point P is at the COR of the RPM, there are no additional accelerations other than gravity, as the vector *P* from point P to the COR is zero. However, as the magnitude of *P* or rotational rate ϖ increases, they are increasingly influential on the resulting *g*-level experienced at point P. Therefore, to track the gravity vector's direction, Eq. (8) is used by combining the global gravity vector of Earth with the rotational matrices defined previously. The angle of the RPM frames can be found at each time step by multiplying the angular velocity ϖ with time *t*. However, because velocity rate or direction is not constant, the angle at time step *n* is calculated and added to the angle at time step *n-1* and so on. The expression is scaled to a unit sphere to fit 1*g* on the axes with *l* indicating points on the trajectory.

$$l = Rx \cdot Ry \cdot \frac{a_g}{9.81} \tag{8}$$

Experimental configuration

Experimental validation was performed with a desktop RPM (Yuri Gravity, Germany) with a platform size of 15.9 cm x 14.6 cm. Inertial motion unit (IMU) sensors with accelerometer sensors were used to assess how rotational rates of the RPM frames and distance from the COR would affect resulting *g*-level at that point. Two IMU sensors were compared: WitMotion "WIT" BWT61CL sensor with a 100 Hz sample frequency (WitMotion ShenZhen Co., Ltd, China), and Delsys "DEL" Trigno Avanti sensor with a 1112 Hz sample frequency (Delsys Inc., Natick, MA, USA). The IMUs were placed on the RPM such that their internal coordinate system aligned with that of the defined kinematics model.

The motion capture system was designed using a 6 degrees-of-freedom optical tracking system with 3 cameras at different locations on the same line (OptiTrack V120 Trio; NaturalPoint Inc., Corvallis, OR, USA). A marker on the bottom of the inner frame was tracked using the camera to record RPM orientation and gravity vector trajectory determination. This had improved performance compared to using multiple markers, which introduced tracking jitters in the motion capture analysis (see Supplementary Videos S1 and S2 online). The same rotation rate was used for both the kinematics model and the RPM to compare their performance.

Results

Validation of the kinematics model

We first compared the performance of the RPM model between the defined kinematics mathematical model above to physical accelerometer data from two inertial measurement unit (IMU) sensors. Table 1 compares the average *g*-level after 40 min of rotation at different rotation rates using the kinematics model, WIT sensor, and DEL sensor. Notably, the RPM performance is closest to a microgravity-analog at lower rotation rates: theoretical and validated *g*-level at $\omega_I = 12$ deg/s and $\omega_O = 24$ deg/s reaches a maximum of 0.0272*g*, whereas $\omega_I = 103$ deg/s and $\omega_O = 120$ deg/s reaches a maximum of 0.0455*g*. Higher rotation rates in an RPM likely contribute to off-nominal accelerations that can reduce the fidelity of microgravity simulation.

Validation of the kinematics model can be seen in Fig. 1C, showing the time-averaged x-direction acceleration of the corner point of an RPM for the kinematics model, DEL sensor, and WIT sensor. At the lowest rotation rates, there is a 10-fold difference (0.010*g*) between the average acceleration detected by the WIT and DEL IMU sensors, whereas the DEL sensor and kinematics model were closely aligned. At higher rotation rates, the model and both sensors were all closely aligned. This may be due to the 10-fold higher sampling frequency of the DEL sensor likely resulting in higher accuracy. Regardless, average absolute variation between the model and sensors is low (0.011*g* WIT, 0.003*g* DEL), indicating physical validation of the kinematics model.

Outer frame (ω_O deg/s)	Inner frame (ω_I deg/s)	Ratio	Model (g)	WIT sensor (g)	DEL sensor (g)
24	12	2.0	0.0023	0.0272	0.0028
24	18	1.5	0.0031	0.0273	0.0027
40	30	1.33	0.0045	0.0155	0.0019
30	40	0.75	0.0059	0.0294	0.0053
75	50	1.5	0.0150	0.0101	0.0107
100	50	2.0	0.0243	0.0178	0.0186
96	73	1.32	0.0263	0.0252	0.0227
50	75	0.67	0.0150	0.0092	0.0123
50	100	0.5	0.0230	0.0176	0.0215
75	100	0.75	0.0281	0.0118	0.0250
120	103	1.17	0.0443	0.0455	0.0411

Table 1. Average G-level after 40 min at different rotation rates.

We then further analyzed the RPM model by comparing its performance to the motion capture recording software. Different ratios of the rotational rates of the frame yield different net trajectory patterns; a representative image of one is shown in Fig. 2A with a highly repetitive trajectory indicated by the narrowness of the model and motion capture recording. The 2D view of the trajectory is also shown on each of the respective xy, xz, or yz coordinate planes. Interestingly, at certain ratios of the rotational rates, the RPM does not reach exact rotational rates and results in drifting of the trajectory path and discordance between the mathematical model and the RPM's physical performance (Supplementary Figure S1). Incorporating the real rotational rates into the mathematical model, instead of the ideal rates, can correct for this drifting and allow for broader, "imprecise" rates in the model itself. This then re-achieves overlap between the model and motion capture recording, as shown in Fig. 2B. Ultimately, the kinematics model was successfully validated then with both the IMU and motion capture systems.

Characterization of the RPM performance

Rotation rates to achieve lowest resulting g-level

To assess the RPM performance as a microgravity-analog, the time-averaged *g*-level of the gravity vector was first analyzed. Figure 3A is a map of the average *g*-level after 25 min of rotation with different combinations of inner and outer frame rotational rates from 1 to 70 deg/s. Table 1 indicates that switching the rotation rates for the inner and outer frames, i.e., the ratio between the frames, can significantly influence microgravity simulation: 30 deg/s ω_I and 40 deg/s ω_O results in *g*-levels approximately double that of 30 deg/s ω_O and 40 deg/s ω_I based on WIT and DEL accelerometer sensor data. To guide selection of rotational rates, Fig. 3B further highlights the lowest *g*-levels achieved for different rotation rates of the outer and inner RPM frames. Critically, at a ratio of 1 between the outer and inner frame rotation rates, i.e., rotation at the same rate, drifting—previously noted during validation of the kinematics model—will significantly affect the resulting *g*-level and prevent microgravity simulation. Optimal microgravity simulation was achieved when the outer frame rotation is at least 30 deg/s, and the inner frame rotation is at least 20 deg/s slower.

Sizing of an experiment on the RPM

The RPM performance as a function of the rotational rates and distance of a point P from the COR is a critical measure for guiding how large of a testbed platform can be placed in the RPM with adequate time-averaged microgravity-simulation. Table 2 shows the resulting time-averaged *g*-level after 30 min for different distances from the COR and at different rotation rates in approximately 10 deg/s intervals. The outer frame was rotated at least 4 deg/s faster than the inner frame to prevent exceedingly high *g*-levels as discussed and shown in Fig. 3B. At slow rotational rates, the resulting *g*-level is minimized even at distances 15 cm from the COR. However, at faster rotational rates such as $\omega_O = 67$ deg/s and $\omega_I = 71$ deg/s, the resulting *g*-level increases significantly as distance from COR increase. At 15 cm from the COR at those rotation rates, a sample experiences over 0.03*g*, representing a significant deviation from time-averaged microgravity simulation.

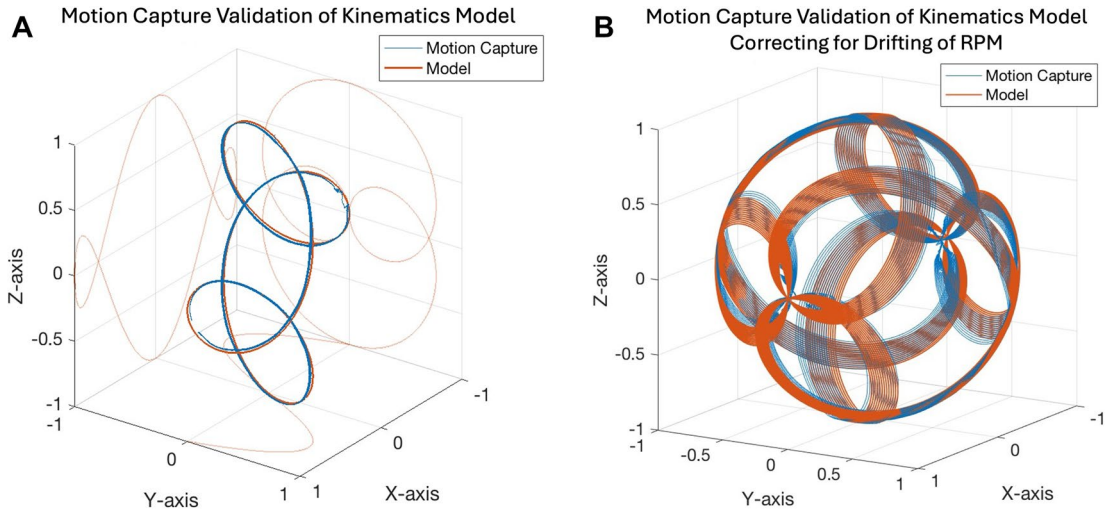


Fig. 2. Drifting of the random positioning machine (RPM) at certain rotational rates. **(A)** Motion capture (blue) trajectory of the inner frame of an RPM during rotation at $\omega_O = 24$ deg/s and $\omega_I = 12$ deg/s compared to kinematics model (orange). The 2D shapes of the model are shown as shadows on the sides. **(B)** Trajectory of the inner frame after 14 min of a run with drifting, shown by the noticeably wider motion capture (blue) lines. This is due to the RPM's inability to maintain precise rotational rates, therefore skewing the model's alignment with its performance. Instead, imprecise rates of $\omega_O = 23.98$ deg/s and $\omega_I = 17.966$ deg/s were maintained. When using those imprecise rotation rates in the model itself, alignment between the two is re-achieved.

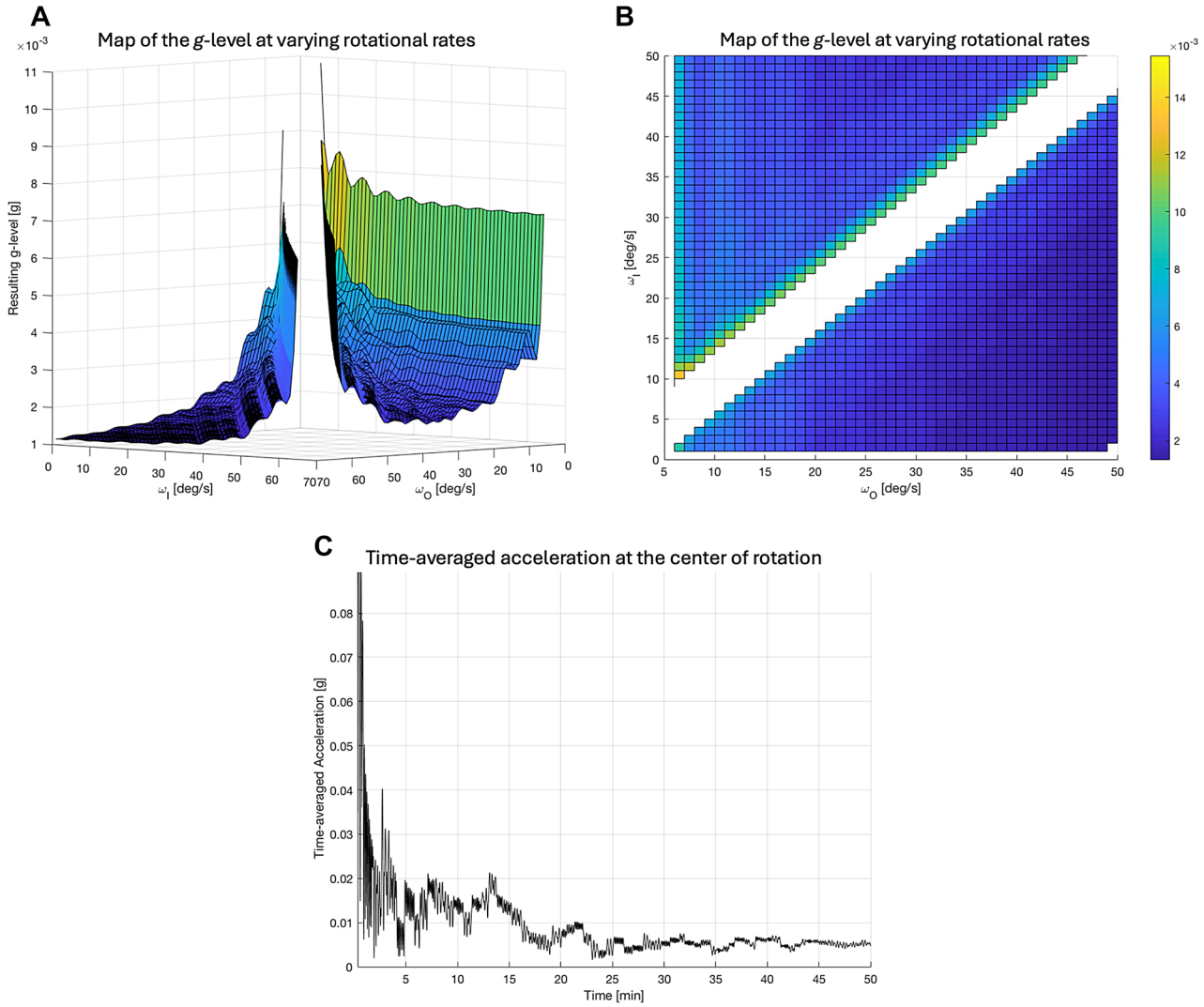


Fig. 3. Characterization of the RPM Performance. **(A)** Mapping the gravitational level resulting from different RPM rotational rates. Cumulative *g*-level after 25 min of varying rotation rate between 1 deg/s and 70 deg/s is shown. The diagonal has been left out to highlight more detail for additional rotational rates. **(B)** Map of the corresponding *g*-level after 25 min with varying the rotation rate of the frames between 1 deg/s and 50 deg/s. *G*-levels from the diagonal and 0 deg/s rotation of either frame have been left out to show more details on the remaining options for rotational rates. The lowest *g*-levels, and therefore optimal microgravity simulation, result when the outer frame rotates at least 30 deg/s and inner frame rotation is at least 20 deg/s slower than the outer frame. **(C)** Time to lowest resulting *g*-level on an RPM. Time-averaged *g*-level is shown through 50 min of RPM rotation. Rate was changed every 30–60 s randomly, between (ω_O ω_I = ±50 ±14; ±48 ±5; ±44 ±16; ±33 ±13; ±20 ±47) deg/s. Average acceleration is minimized after 25 min of RPM rotation, never reaching above 0.01 g after that point.

Time to lowest resulting g-level

We then sought to determine how long an RPM experiment must run before achieving the lowest possible *g*-level, as the RPM would only be suitable for experiments longer than that. Time-averaged *g*-level of the RPM is shown over the course of 50 min of random rotation in Fig. 3C. Notably, the *g*-level significantly decreases over time but in a non-monotonic fashion; *g*-levels consistently under 0.01*g* are not achieved until at least 25 min of random RPM rotation.

Using the RPM across the gravity continuum

As shown, faster rotational rates and higher distances from the COR both introduce off-nominal accelerations that result in higher *g*-levels. For most RPM experiments, this is not ideal as lowest possible *g*-levels are required for microgravity simulation. However, with exploration-class missions planned to the Moon and Mars, understanding biological and physical responses to partial gravity environments is increasingly critical[27]. Figure 4A validates the RPM performance as a simulator for lunar 0.166*g* partial gravity, as the time-averaged accelerations can be specifically shown to be 0*g* in the x- and z- directions but 0.17*g* in the y-direction.

Average gravity level (g) at varying speeds and distances								
Rotation (deg/s)		Distance From Center of Rotation (cm)						
ω_O	ω_I	0	0.1	0.5	1	5	10	15
9	13	0.0031	0.0000	0.0000	0.0000	−0.0001	−0.0002	−0.0003
23	18	0.0019	0.0000	0.0000	0.0001	0.0006	0.0014	0.0022
31	27	0.0019	0.0000	0.0001	0.0002	0.0014	0.0032	0.0050
39	43	0.0020	0.0000	−0.0002	−0.0003	0.0011	0.0049	0.0089
53	49	0.0018	0.0001	0.0004	0.0009	0.0054	0.0112	0.0171
63	58	0.0017	0.0001	0.0007	0.0014	0.0079	0.0162	0.0245
67	71	0.0020	−0.0002	−0.0004	−0.0001	0.0080	0.0192	0.0305

Table 2. Average g-level after 30 min at different distances from the center of rotation (COR) for different rotation rates. The values away from the COR (> 0 cm) are the increase in g-level compared to the COR (0 cm), and the distance is taken strictly in the positive x-direction.

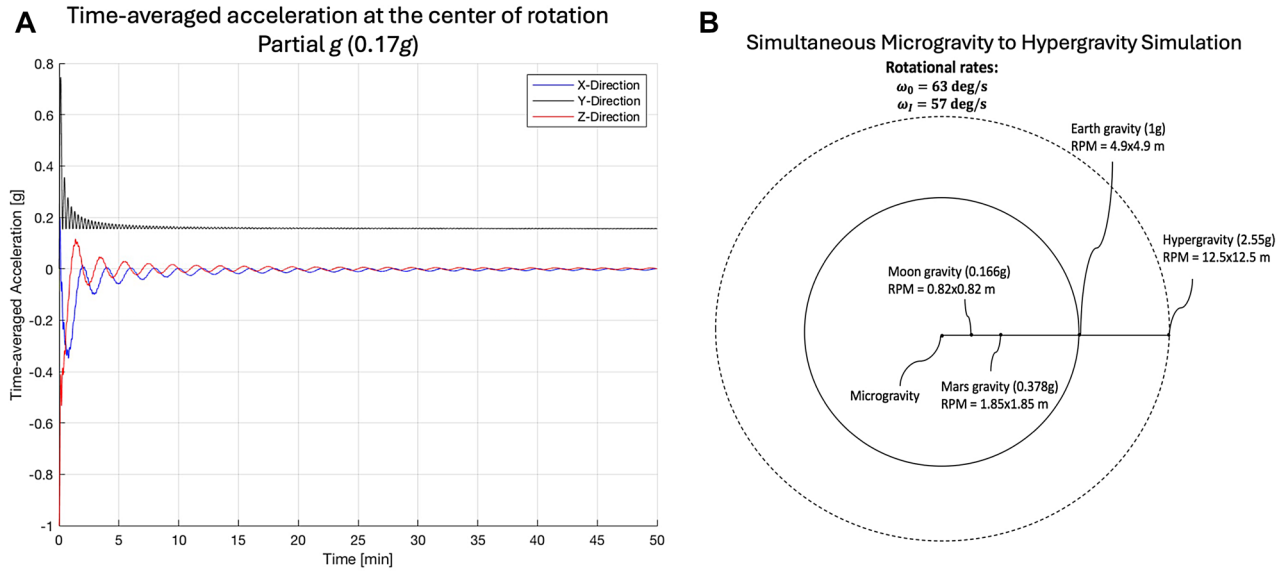


Fig. 4. RPM design for simulation across the gravitational spectrum. (**A**) Demonstration of an RPM for lunar partial gravity (0.166g) validating that the RPM can indeed be used for partial gravity simulation in the y-direction. (**B**) Microgravity to hypergravity can be achieved within the same RPM depending on rotation rates and a sample's distance from the center of rotation. This schematic uses constant rotation rates at ω_O = 63 deg/s and ω_I = 57 deg/s, but the same determination of RPM size can be performed at any rotational speeds.

Further, it is possible to intentionally utilize the off-nominal accelerations and higher resulting g-levels to simulate a range of g-levels within the same RPM experiment. One RPM platform to simulate diverse g-forces from microgravity to partial gravity, and even hypergravity, can enable efficient investigations into gravitational dose-responses for various systems. The specific design of such an RPM platform will significantly depend on desired rotation rates, as faster rotation rates can use smaller RPM's to achieve the same g-level. Using sample rotation rates of ω_O = 63 deg/s and ω_I = 57 deg/s, Fig. 4B shows how a larger RPM can achieve microgravity at the COR, lunar 0.166g gravity 0.82 m from the COR, Martian 0.378g gravity 1.85 m from the COR, and even 2.5g hypergravity if a sample is placed 12.5 m from the COR. Slower rotation rates would require an even larger RPM: halving the ω_O and ω_I would require a 4x-larger RPM (Table 3).

Pole bias

For accurate microgravity simulation regardless of g-level, the RPM should rotate a platform and its respective gravity vector through all possible orientations. Therefore, the gravity vector should be uniformly distributed around the entire unit sphere and not concentrated at any particular orientations. However, we have found that high rotational rates contribute to an observation of "pole bias": when the outer frame is aligned vertically, which occurs twice during each rotation, the gravity vector is aligned with a sample and generates two poles. At higher rotation rates, this pole is generated more frequently over time, and therefore results in a pole bias for the RPM. The gravity vector trajectory path of the RPM has higher concentrations at pole coordinates (0,1,0) and (0,−1,0) on the unit sphere for random rotation rates between 2 and − 2 revolutions/min (Fig. 5A). This pole bias persists

RPM Sizing (m) to achieve different gravity levels					
Rotation (deg/s)		Gravity Level (*g*)			
ω_O	ω_I	Moon (0.166*g*)	Mars (0.378*g*)	Earth (1*g*)	Hypergravity (2.5*g*)
63	57	0.82 m	1.85 m	4.9 m	12.5 m
33	27	3.2 m	7.2 m	20 m	48 m

Table 3. RPM sizing across the gravity continuum for different rotation rates.

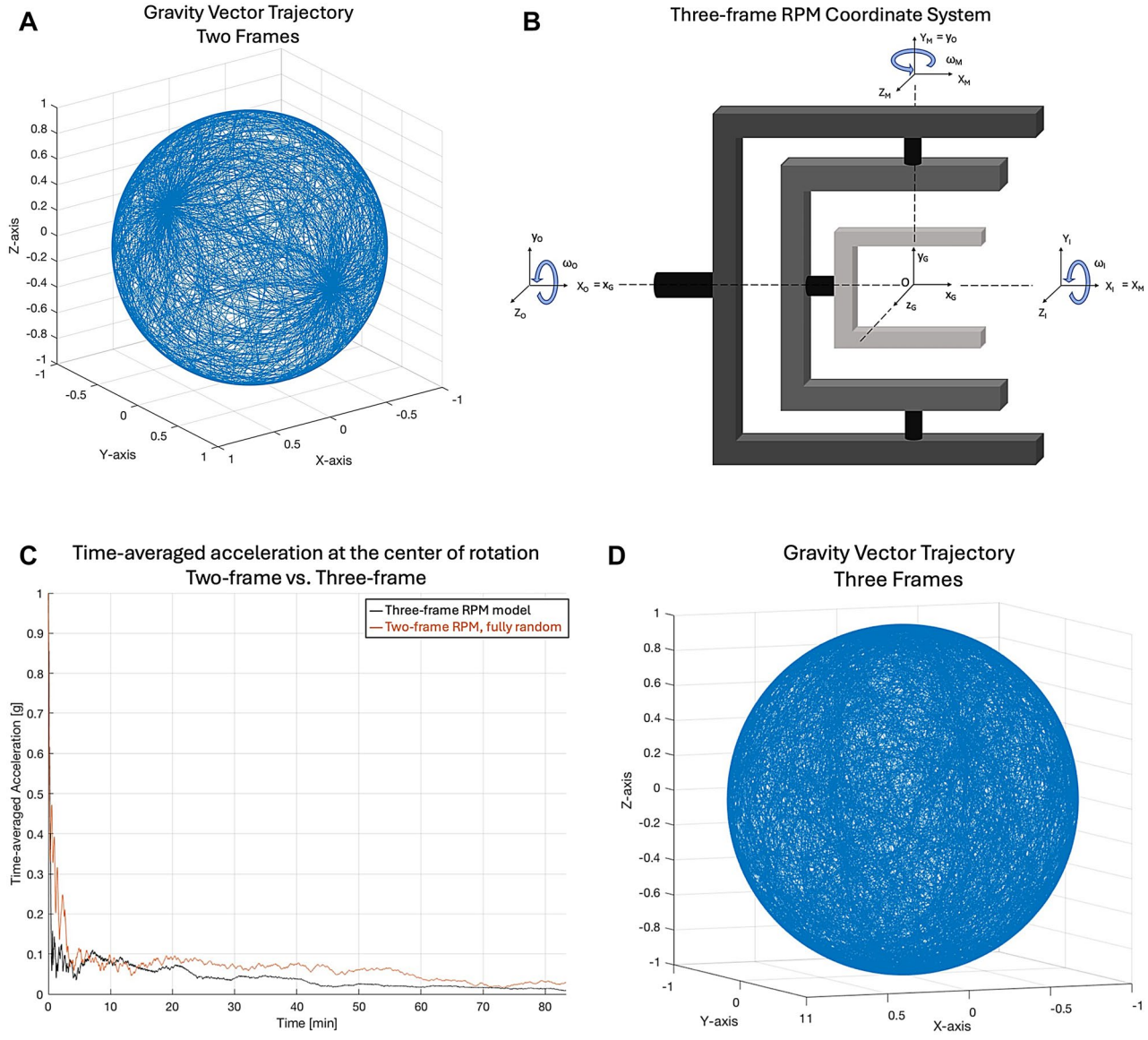


Fig. 5. Pole bias in an RPM can be minimized with a three-frame design. (**A**) Pole bias in the cumulative trajectory of the gravity vector of a two-frame RPM after 3 h with rotation rates between 2 and −2 revolutions/min for both frames. Direction was randomly changed every 30–60 s. RPM was allowed 2 s to reach the new rotational rate. (**B**) Reference model and schematic diagram of a three-frame RPM with 3 distinct coordinate systems. (**C**) Time-averaged acceleration experienced by a point at the center of rotation (COR) for a three-frame RPM model (black) and two-frame RPM (orange) under fully random rotation for all frames. (**D**) Trajectory path of the three-frame RPM model after 83 min of rotation. Outer frame was constantly rotated at 27 deg/s while the inner and middle frame randomly rotate between selected rotation rates. The three-frame design prevents pole bias while still covering the entire unit sphere of the gravity trajectory paths.

with alternative control algorithms such as constant rotation rates that are randomly inverted or sinusoidal rotation of the outer frame so that it only rarely moves through the vertical direction (Supplementary Figure S2). Sinusoidal rotation of the outer frame with the inner frame rotating either constantly or randomly (between 2 and −2 revolutions/min) does prevent pole bias, but the gravity trajectory path does not cover the entire unit sphere and therefore does not represent an adequate microgravity-simulation (Supplementary Figure S3). Both constraints, (1) complete coverage of all gravity vector directions and (2) prevention of pole bias, are critical for an accurate time-averaged microgravity simulation.

However, in lieu of developing advanced control algorithms, a three-frame RPM may address these constraints. Given the validation of the two-frame kinematics model with using physical RPM performance data, we then developed a kinematics model defined similarly for a three-frame RPM (Supplementary Equations S4-S15). The reference frame and coordinate system are shown in Fig. 5B. The time-averaged acceleration experienced by a sample at the COR of a three-frame RPM under fully random rotation is shown in Fig. 5C; after 20 min of rotation, the three-frame RPM has superior time-averaged microgravity-simulation compared to a two-frame RPM under similarly fully random rotation for all frames. A cumulative trajectory path for a three-frame RPM's gravity vector is shown in Fig. 5D, where the outer frame rotates constantly, and the middle and inner frames randomly rotate. Notably, a three-frame RPM under this relatively simple control algorithm will rotate a platform through all gravitational orientations (and cover the entire gravity trajectory path on a unit sphere) without any pole bias.

Discussion

As we embark on the ambitious goals of interplanetary spaceflight missions, understanding biological, physical, and material science phenomena in microgravity and partial gravity is a critical task. RPMs are widely used by space scientists to investigate these effects in a more robust, operationally efficient manner. Functionally, an RPM rotates around two frames at random rotation speeds and directions that change at random time intervals[18,19]. Over time, the direction of the gravity vector for a sample will cover all possible orientations and therefore represent an adequate analog for microgravity[18,19]. This holds true for a singular point at the COR, but additional forces arise farther from the COR such as centripetal, Coriolis, and Euler accelerations that may reduce this simulation fidelity; it has been unknown how far from the COR adequate hypogravity can still be achieved depending on rotational velocity. Therefore, we both defined a complete mathematical model for the RPM and experimentally validated it to determine these guidelines for optimized design of spaceflight experiments.

We first developed a kinematic model of the RPM to rigorously define the acceleration (Eq. 7) and gravitational level (Eq. 8) experienced by a sample at the COR. In comparing this ideal mathematical model to the RPM's true performance with accelerometer sensors and motion capture data (Fig. 1), we successfully validated the RPM as a microgravity-analog within certain experimental parameters. This aligns with the RPM's marked improvement as a microgravity simulator compared to traditional 2D or 3D clinostats[14].

We characterized the *g*-level experienced by a sample at the COR of an RPM using different inner-frame and outer-frame rotation rates. Drifting deviation of the RPM performance from the kinematics model occurred when both frames rotated at the same rate and should therefore be avoided. Slower rotation rates led to minimized off-nominal accelerations, though the relative ratios of inner frame and outer-frame rotation rates were also significant factors in the performance of the RPM as a microgravity-simulation (Table 1). Critically, the rotation rates of the two frames should be ≥ 10 deg/s different from one another. Minimum *g*-levels, and therefore optimal time-averaged microgravity simulation, were achieved when the outer frame rotated at a rate ≥ 30 deg/s and the inner frame rotated at least 20 deg/s slower (Fig. 3). The use of the RPM should likely be limited to experiments that can fit within these rotational rate parameters, as off-nominal accelerations and forces such as shear stress can significantly influence biological sample behavior[28,29].

The kinematics model additionally demonstrates how, as a sample's distance from the COR increases, the centripetal, Coriolis, and Euler accelerations have increasing contributions that influence the resulting *g*-level in conjunction with rotation rate (Table 2). At slower rotation rates, the sample can be placed 15 cm from the COR while still experiencing an adequate time-averaged microgravity-analog *g*-level. However, at faster rotation rates greater than 50 deg/s, samples should be less than 10 cm from the COR to prevent unacceptably large accelerations and resulting *g*-levels. There is a clear interplay between rotation rate and sample distance from an RPM COR that affects a sample's experienced *g*-level over time and fidelity of microgravity simulation.

Time-averaged *g*-level experienced does decrease non-monotonically with longer experimental durations; consistently low *g*-levels under 0.01*g* are achieved after 25 min of rotation, supporting the use of an RPM for experiments that are ideally at least that long (Fig. 3). For example, combustion studies on the order of milliseconds are therefore not suited for an RPM, whereas slower life sciences—e.g., plant growth or cellular behavior—assays are appropriate[11,30,31]. Given the development of incubator-compatible RPMs to facilitate such studies, growing interest in investigating the effects of combined microgravity and radiation[1,3]may also translate to the development of more advanced radiation-compatible RPMs as well. Regardless, understanding optimal rotation rates, experiment duration, and sample size as outlined herein should guide thoughtful design of successful microgravity-analog studies using the RPM.

Partial-gravity analog studies are also of increasing interest, considering commercial and federal plans for missions to the Moon and Mars. Transitioning our gravitational paradigm from a 0*g* versus 1*g* binary to a 0*g* through 1*g* continuum will enable deeper understanding of biological and physical phenomena at lunar (0.166*g*) and Martian (0.378*g*) gravity[11,32]. We successfully validated the RPM performance for partial gravity experiments and further proposed the design for larger RPM platforms that can facilitate the simulation of *g*-levels from microgravity to hypergravity in a single run (Fig. 4). Although individual hypergravity simulation can be easily achieved using established centrifuge equipment, this larger RPM platform would have the particular advantage of achieving simultaneous microgravity through hypergravity simulation. This may enable novel investigations

through such simultaneous exposure, larger experimental volumes, more sophisticated equipment design, and improved operational efficiency.

Finally, we characterized "pole bias" as a critical limitation in the performance of a two-frame RPM's microgravity simulation. In addition to minimizing time-averaged *g*-level, the direction of the gravity vector and its trajectory path should be uniformly distributed around the unit sphere. However, we found that the for a two-frame RPM, the gravity vector is aligned vertically with a sample twice during each rotation; over time, the gravity vector therefore becomes more highly concentrated at those vertical poles and therefore is not uniformly distributed over the unit sphere (Fig. 5). This occurs more frequently with higher rotation rates and persists even with alternative control algorithms. Therefore, we developed a kinematics model for a three-frame RPM and demonstrated its potential to both minimize time-averaged *g*-level and rotate a platform without any pole bias. An implicit assumption within this is that the three frames must achieve true independent, random rotation; correlations between the three frames' rotations may re-manifest pole bias regardless. Future work, enabled by this design's characterization herein, should therefore focus on statistically verifying the randomness of rotational control algorithms to ensure pole bias does not reappear depending on initial conditions or trajectories. A three-frame RPM may nonetheless represent a more feasible mechanical solution to the pole bias limitation, in lieu of developing advanced control algorithms.

The RPM therefore represents a now-validated rotational platform that can be used as a microgravity-analog for spaceflight experiments. Using this set of experimental guidelines, spaceflight scientists can design studies with optimized microgravity simulation performance. We further validated the RPM's behavior as a partial gravity analog, which will be increasingly valuable as we continue to work towards enabling long-duration exploration-class lunar and Martian missions. Characterizing biological phenomena across the gravitational continuum can be efficiently enabled using an RPM platform, transforming our ability to safely explore the next frontiers in spaceflight sciences.

Data availability

All data generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Author contributions
All experimental work was completed at Massachusetts Institute of Technology. Conception and design (LGP). Acquisition, analysis, or interpretation of data for this manuscript (AW, LB, LGP, JCGP). Drafting of the work (AW, LB) and revising it critically for intellectual content (AW, LB, LGP, JCGP). All authors approved of the final version of the manuscript and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Declarations

Competing interests
The authors declare no competing interests.

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